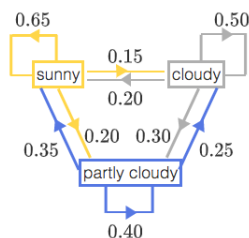
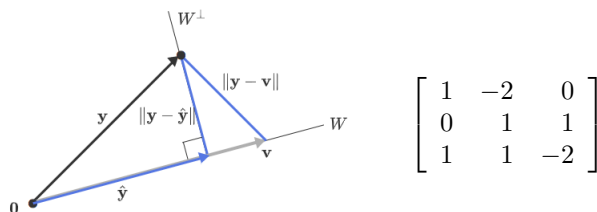


MAT 315: Linear Algebra

Augsburg University – Spring 2025, §A



$$\begin{cases} x_1 - 2x_2 & = 2 \\ & x_2 + x_3 = -1 \\ x_1 + x_2 - 2x_3 & = 4 \end{cases}$$



General Information

Class meets MWF 1:50-3:00 PM in Hagfors 151. Our last quizzes are Wed May 7 from 3:00-5:00 PM, which is our assigned final exam time. Classes meet in person but might occasionally switch to Zoom.

Professor Dr. Matthew Haines, haines@augsborg.edu, Hagfors 181. You are welcome to call me Professor Haines, Dr. Haines, or just Professor.

Office/help hours No appointment is necessary to come get help during my office hours – just drop in! My office hours are in my office (HC 181) or the study area across from my office: **MWF 11:00 AM-12:00 PM**, and **Tu 5:00-5:45 PM**, and **Th 2:00-3:00 PM**. Feel free to email me with questions, and I will reply typically within a day. While I don't have a regularly scheduled Zoom help session, feel free to request at least a day in advance to meet via Zoom if you won't be on campus. I will give you advance notice if I have to change my office hours. If I am not available try working with a classmate, or e-mail me.

Textbook & Technology

- Our free textbook is *Understanding Linear Algebra* by David Austin; the link is posted to Moodle.
- You must buy Edfinity access for the online homework, either from Augsburg's bookstore, or directly online; instructions are posted to Moodle.
- You will need regular access to the Internet to read the textbook, to complete homework problems Edfinity, and to access resources on Moodle. If you do not have reliable Internet at home, contact the Info/Tech Desk at Lindell Library for help. (They can also connect you with a laptop or noise-cancelling headphones to help you work from home.)
- You will need regular access to a device that scans written work; any cellphone with a camera has this functionality. The on-campus printers also scan. Alternatively, you can use a writable electronic surface.
- Many days it will be useful to have a laptop in class but you will be working with a classmate, so often it is possible to share.

Prerequisites MAT 145 Calculus I and at least one of the following courses: MAT 245/255 Calculus III/Multivariable Calculus, MAT 271/302 Discrete Mathematical Structures, or DST 234 Introduction to Data Science (and R). In any prerequisite course, you must have earned a grade of C- or higher (or P). If you received B- or lower in the prerequisite or any college mathematics course, please meet with me early in the term to help you plan for success.

Standards met This course, with a grade of C- (or P) or higher

- Is required for majors in Data Science, Mathematics (BA and BS degrees), and Mathematical Economics major and for Augsburg's Education Licensure programs in secondary (grades 5-12) mathematics;
- Is recommended for the Computer Science major (BS degree);
- Counts towards the minor in Mathematics; and
- Is a prerequisite for MAT 314 Abstract Algebra and DST 475 Machine Learning.

Students who have completed MAT 246 (or its transfer equivalents) may not register for credit. If you have taken a full-semester linear algebra course at another institution, please meet with me right away.

Course topics

- **Systems of linear equations:** modeling with linear systems, geometry of linear systems, representing infinitely many solutions, and solving by Gaussian elimination.
- **Matrix representation:** row reduction, matrix operations, properties, determinants, eigenvalues, eigenvectors, and diagonalization;
- **Vector linear combinations:** vector operations, linear independence, spanning, basis, dimension, coordinates, orthogonality, and orthogonal projection;
- **Linear transformations in $\mathbb{R}^n$** and geometric transformations of $\mathbb{R}^2$ and $\mathbb{R}^3$;
- **Abstract vector spaces** including subspaces and linearity.
- **Applications:** Introduction to applications such as network flow models, mixing problems, economics models, linear programming, dynamical systems, Markov chains, computer animation, image compression, Google's Page Rank algorithm, k -means clustering, and linear regression.
- **Introduction to computational linear algebra:** Use of computer functions and programming (in SAGE) to perform computations and limitations.

Student learning objectives

- Learn basic skills and concepts of linear algebra and linear geometry. There are 20 Learning Standards (LS) used to measure this objective. The list is posted to Moodle.
- Appreciate the utility of linear algebra and be able to model applied situations using linear structures.
- Be able to translate between representations and to recast a problem posed in one representation into a different representation where the problem is easier to solve.
- Strengthen your mathematical reasoning skills including geometric visualization; generalizing concepts and abstraction; and construction of examples and counterexamples.
- Be introduced to the theoretical structure of mathematics including the role of definitions, examples, theorems, and reading and writing elementary proofs.

How to learn in this class (or tips for getting an A)

Being in class to have this opportunity to learn with me and classmates is very important. We will work on learning the new material, primarily by working in pairs on activities. I will introduce new ideas, summarize what we are learning in the activities, discuss important course policies, and remind you of important deadlines. You will not want to miss class! I understand that at times you might occasionally have a serious situation that leads to your missing class. Please try to avoid missing class in less serious situations. We will use all of class time so please arrive on time and stay for the full session.

Building a supportive classroom where you feel welcome is a priority for me and I ask your help making sure all students are welcome and included. I have various course policies to support this work. I encourage you to embrace these practices, even if they seem a bit new to you, as I believe they will help you succeed.

Your course grade is based on three categories: engagement, written homework, and quizzes. The first way to get engagement points is to attend class. If you are in class from start to finish and paying attention, you get 5 engagement points. If you arrive late, step out during class, leave early you usually get 4 points. If you miss class or are not participating you get 0 points. You can earn back points up to a max of 4 points by writing up the activities and giving it to me in person or via email within a week of the class. Don't forget to submit activities for any day you miss.

There are two type of homework problems: online and written. Online homework is completed through Edfinity. Be sure to actually work the problems – avoid any sort of guessing because that does not help you learn. Follow instructions as to whether or not to use computational technology. The quizzes won't allow technology, so some practice without is important.

Edfinity grades your problem immediately and you can redo any problem until you answer it correctly. The highest grade stands so you can even go back and try it again when studying for quizzes without risk of accidentally lowering your grade. Each Edfinity assignment is worth up to 10 engagement points (ceiling of percentage/10) with an 11th extra credit point if you get 100%. You can only get 100% if you complete the assignment on time, which is usually due MWF at 11:59 PM. Make it your habit to complete the online homework well in advance of the due time both so you can possibly earn extra credit but also so you can get help if you need it. If you do not complete an assignment on time, you can do it late but there is a penalty. You earn 100% of points earned before the due date and 80% of points earned after the due date. All Edfinity links close at the start of our assigned final exam time.

The written homework is posted to Moodle and submitted on Moodle. Some of the problems on written homework reinforce key ideas and give you a chance to get feedback on the concepts and skills tested on our quizzes, especially how you are explaining your work. Other problems show you applications or interesting ideas that extend beyond what is on the quizzes. Written homework often ask for you to explain and show your work, which is also good practice for the quizzes (and life). While many problems are relatively straightforward, I like to include many more challenging problems too, so do not be surprised if you get stuck or are not able to complete every problem fully. That is okay and you should just turn in whatever you can do.

Written homework is typically due Mondays and Wednesdays by 11:59 PM, although they are accepted through 11:59 AM (just before noon) the following day without penalty. At noon that day the solutions will appear on Moodle. You will then have a few days to grade your original work, revise and correct any problems based on the solutions, and grade your revised work. The revised/graded version of the assignments are typically due Fridays by 11:59 PM, although they are accepted up until the Moodle link closes one week later without penalty. Later submissions are by special arrangement only. Grading your own assignments is the best way to reflect on your own work and learn from any mistakes you might have made. Try to have fun with it. You are welcome to work with classmates on the grading or ask me questions about it.

There are two practices that will save you time and increase how well you learn the material. First, get in the habit of starting to starting both the online and written homework right away after class, at least to fill in whatever you can from what we did in class. Second, try to finish the homework before our next class session, even though it is not due until later. That way you can meet with classmates to compare work and come see me during office hours (or email me) to get your questions answered well before the last minute.

You are welcome to work with classmates on assignments or to ask me questions during office hours or via email. If you work with a classmate, write up the final solution yourself to make sure you understand it. Please do not copy from classmates, look for solutions online, or otherwise compromise the integrity of your work. Plus, those strategies might get an assignment “done” but they will not help you learn the material.

There are no exams and no final exam in this course. Instead, we have weekly learning standards (LS) quizzes, usually two LS per week. These quizzes are closed book and individual – time to show me what you have learned. No reference material, computational technology, or collaboration is allowed on the quizzes, unless we have alternative accommodations in place.

These quizzes cover the course learning standards (LS). Each quiz is marked “Met” (full credit) if it is 100% correct or “Not Yet” (no credit) otherwise. You can retake quizzes several times, including a last-chance on up to six quizzes during our scheduled final exam time. See Learning Standards and Quizzes for lots more detail. Make it a goal to be prepared for quizzes to try to meet the LS on the first try (or at least the second).

Study for the quizzes by reviewing class notes and your completed assignments. Also work on the practice quizzes which are posted to Moodle. The practice quizzes are due by the start of class the first time a quiz is offered and are worth 5 engagement points. Late practice quizzes are accepted until the “last chance” date and can earn up to 4 points. Although solutions to the practice quizzes are posted to Moodle, try doing the quizzes first without consulting the solutions but then be sure to check your work against the solutions to identify areas that need further study.

How to submit, revise, and grade your written homework

There are two choices for how to write up your solutions to written homework.

- Print the PDF, write solutions by hand on that paper, scan (not photo) the pages into a single PDF, and upload to Moodle. Note that you can print free on campus (and if you run out of prints ask for more free prints). Please check that your scans are easily readable and have the pages in order. -or-
- Open the PDF in an annotation software, use a writable surface and stylus to write solutions by hand, Export to PDF (to capture your work), and upload to Moodle.

In either case your solutions must be hand-written unless we have an alternative accommodation in place.

The first step in grading each problem is to read the posted solutions and evaluate your work using the following scale. Note that each problem is graded (not each part).

- 5** Correct, complete, and clearly written solution.
- 4** Nearly correct, complete, and clearly written solution.
- 3** Some correct, relevant progress toward a solution OR correct answer with no work.
- 2** Work shows understanding of the problem but little or no progress toward a solution.
- 1** Attempted but does not show understanding of the problem.
- 0** Not attempted.

For scores of 0, 1, 2, or 5 no comments are required. For scores of 3 or 4, include some brief comment about what is incorrect, missing, or unclear.

Next, use the posted solutions and other allowed resources to revise your solution. Allowed resources are class notes, the textbook, work you did on online or written homework, anything posted to Moodle for this class, classmates, me, and nothing else. Please write your revisions in a bright color (pink, orange, purple) that you did not use for your original work and make sure the color is distinguishable from your original work on your scans. If you already had a 5, then there is nothing to do. If you had a 3 or 4, please keep work that was correct and add/fix the rest. If you had a 0, 1, or 2 you should redo the problem and it is okay to copy my solutions, hopefully with them making sense to you.

Then, grade your revised work using the same scale. Typically you will have a 5 for this second grade. Note that if you don't want to correct your original work (or run out of time), then you will have the grade on that problem again.

At this point in the process each problem should have score written to the left of the work in the format **3+5**. Please keep in that format (versus writing 8). Note that if you did not submit the work by the original due date that your first score is a 0, but **0+5** is still possible. I strongly encourage you to submit whatever you have completed in time for it to count towards that first score.

Last, calculate your grade on the entire assignment using the following formula, recalling the ceiling function rounds up to the nearest integer:

$$\left\lceil \frac{\text{Total Points Earned}}{\text{Number of Problems}} \right\rceil$$

Please write the entire formula and grade on the top of your first page. For example, if you earned a total of 44 points on 6 problems, you would write $\left\lceil \frac{44}{6} \right\rceil = \lceil 7.33 \rceil = 8$.

If you submit the original assignment but do not grade it, I will assign a grade of at most 5 points without feedback on individual problems. I reserve the right to limit how often I mark ungraded work. At times I will review your assignments and grading and reserve the right to change the grade if there is a major discrepancy. You are welcome to ask me to review an assignment during office hours with you.

What I expect of you (and course policies you should know)

- Check your Augsburg email account regularly and access our Moodle site as needed.
- Attend and participate in every class. Be on time and prepared for every class. Arrange to stay for the entire class period. If you must come late or leave early on special occasion, please let me know in advance. Avoid scheduling appointments that overlap class time. If you have an occasional excused absence, send me an email as soon as possible so that I know you are okay.
- Pay attention during class. The only thing you should be reading, viewing, writing, or discussing during class is the current material. You may use electronic devices that lay flat on the table (e.g. cellphone or tablet) for taking notes including photographs of the whiteboards, calculations, or approved accommodations, but not for other purposes during class. Laptops are usually not allowed for note-taking (unless they lay flat or you have a documented disability-based need.) When we use laptops to access Activities or for computation during class, that is the only thing you should be looking at on your laptop.
- Do all of the assigned work. Plan on spending 6-9 hours a week outside of class time working on the course. Try scheduling at least 3 hours of work time (perhaps 1 hour + 2 hours) between every pair of class sessions. Your study time should include reviewing class notes and activities, reading the textbook, completing online homework and written homework, grading written homework, completing practice quizzes, and studying for quizzes.
- Submit work when due. You may submit work for as long as the link on Edfinity or Moodle allows, subject to late penalties. I reserve the right to limit how many and which assignments are eligible for late submission.
- Get help when you need it. I know it is hard to ask for help sometimes, but I really enjoy helping students to learn so please reach out to me for help as you need it. You would be wise to arrange to meet with a classmate outside of class to compare work as well. Do not wait until the last minute to get help or meet with classmates. If you did not do well in previous college mathematics courses, or it has been a few years since you took those courses, you should make every effort to arrange your schedule to be able to attend office hours.
- Take quizzes when scheduled. Since you are already allowed to retake quizzes at later quiz days, I do not normally give “make-up” quizzes if you miss class on a quiz day. If you have a potential conflict with our assigned final exam date, discuss the options with me well in advance. If you fall far behind on the quizzes, talk with me – I might let you take a few during office hours to help catch up.
- Work with others in the class when asked to do so. Work alone when asked to do so. Follow all academic honesty rules. To be clear, you are encouraged to share ideas, check answers, and compare solutions or explanations with classmates, but any work you submit must be “your own work” as defined by the Augsburg University Academic Honesty Policy. Direct copying is forbidden. You may not get solutions from additional texts or resources or from people not currently in the class without my written permission. In particular, you are not allowed to work with tutors or other professors without my written permission. The consequence of a violation of this policy is anywhere from a 0% on the work to an F in the course. I am also required to notify the Provost’s office who may impose further consequences. Please protect your reputation and uphold your integrity by committing to being academically honest throughout this course. If you ever unsure if something is allowed, please ask me.
- Check Moodle gradebook and your Edfinity account regularly and email me if you have any questions about your grades. I have to manually enter Edfinity grades into Moodle, so if you do late work and don’t see it entered in Moodle, feel free to email me.
- Keep me up-to-date on how the class is going for you. Be sure to come in and talk with me if I email you or send an Academic Alert so that we can get you back on track.
- Let me know of any special needs or concerns you have. If you need accommodations because of a disability, please contact the CLASS Office (612) 330-1053 or class@augsborg.edu.
- Creating a classroom environment that welcomes and values all students is very important to me. I ask your assistance in helping support one another by following classroom rules. If you feel unwelcome or unvalued at any time, or have any suggestions for improvement, please let me know so I can try to help. Please do not be shy about telling me what is or is not working for you.

- Complete any in-class and online course evaluation forms.
- Understand and comply with university-wide measures to reduce COVID-19 and influenza transmission. I greatly appreciate it if you stay away or wear a mask if you are sick and might be contagious, even if it's "just" a cold. You are welcome to wear a mask at any time.
- As with all Augsburg classes, you should understand and comply with all University policies described in the University catalog and the Student Guide: <http://inside.augsburg.edu/studentaffairs/studentguide/>. In addition, be aware of the statement on FERPA and privacy as well as on Title IX Resources.

How I calculate your course grade

Your course grade is based on three categories.

- **Engagement:** Class activities (5 points/class), online homework on Edfinity (10 points/class), practice quizzes (5-10 points/week), and occasional extra credit opportunities.
- **Written homework** (10 points/written homework)
- **Quizzes** (1 point/quiz)

The percentage of points you earn in each category are translated to a letter grade as follows.

Minimum percentage required in the category	94%	84%	74%	64%	0%
Letter grade for the category	A	B	C	D	F

At the end of the course, I compute your course grade according to the following specifications.

- A requires an A in each category.
- B requires a B or better in each category.
- C requires a C or better in each category.
- D requires a D or better in each category.

Note that Moodle gradebook will not show +/- grades, but I will assign +/- as follows:

- I assign A-, B-, or C- if you have the required grade in two categories and the third category is within 4% points of the cutoff. Example: if you have A in Engagement, A on Quizzes, and 90% on Written Homework (B, within 4% points of an A), then your course grade is A-.
- I assign B+ or C+ if you have the required grade in each category and an A in at least one category. Example: if you have B on Engagement, A on Quizzes, and C on Written Homework, then your course grade is C+.

I reserve the right to lower your Course Grade if your attendance, punctuality, participation, preparation, or behavior falls well below expectations, listed in this syllabus. If during the term you are not progressing towards a passing grade or if your attendance, participation, preparation, or behavior is falling well below expectations, I will usually send an Academic Alert.

I reserve the right to lower your course grade if you violate the Academic Honesty Policy in this course. If that happens, I will notify you in writing. Also, I am required to report any violation of the Academic Honesty Policy. Again, if you are ever in a situation where you are not sure if something is allowed, please ask!

You are welcome to take this course P/LP/N (Pass/LowPass/No Credit) but you should check with your faculty mentor first because there is a limit to the number of P/N courses overall, towards a major, and towards a minor. Some students like this option because the grades of P, LP, and N do not count into your Grade Point Average (GPA). The cutoff for P is the same as for C- and the cutoff for LP is the same as for D.

To get credit for the course you need a grade of D or higher, P, or LP.

For this course to count towards any major or minor offered by the Department of Mathematics, Statistics, and Computer Science (MSCS) or for Education licensure, you must earn a grade of C- or higher (or P).

For this course to serve as a prerequisite for any CSC, DST, or MAT course, you must earn a C- or higher (or P). While a C- or higher (or P) satisfies prerequisites, any grade below a B usually means you will have significant challenges in MAT 314 Abstract Algebra and DST 475 Machine Learning. So, if you're planning to take either of those courses you should work towards a B or higher in this course to be adequately prepared to go on.

MAT 315 Course Calendar V 1.0

Spring 2025

Monday	Wednesday	Friday
January 20 No classes: Martin Luther King Jr Day	January 22 Modeling with linear systems (1.1A) <i>Applications: linear systems</i>	January 24 Solutions of systems (1.1B)
January 27 Row operations and RREF (1.2A) <i>Applications: in=out models</i>	January 29 The formal Gaussian Elimination algorithm (1.2B)	January 31 An introduction to computation in Linear Algebra with Sage (1.3) Quiz LS 1
February 3 Pivots & their influence on solutions spaces (1.4)	February 5 Introduction to vectors (2.1A/6.1) <i>Application: mixture models</i>	February 7 Dot products, orthogonality, and angles (6.1) <i>Application: similarity of data vectors</i> Quiz LS 2
February 10 Linear combinations of vectors (2.1B) <i>Application: mixture models revisited</i>	February 12 Matrix-vector product and writing solutions as linear combs (2.2A)	February 14 The span of a set of vectors (2.3A) Quiz LS 3-4
February 17 Spanning sets for $\mathbb{R}^n$ (2.3B) <i>Application: RGB and color models</i>	February 19 Linear dependence and independence (2.4A)	February 21 Homogeneous systems (2.4B) Quiz LS 5-6
February 24 Matrix operations (2.2B)	February 26 Linear transformations (2.5A)	February 28 <i>Application: discrete dynamical systems</i> (2.5B) Quiz LS 7-8
March 3 Geometry of linear transformations (2.6)	March 5 Invertible linear transformations (3.1A)	March 7 Invertibility (3.1B) Quiz LS 9 (last chance LS 1-6)
March 10 Matrix transpose (6.2A) and Orthogonal sets, matrices, and linear transformations (6.3A)	March 12 Determinants (3.4)	March 14 Basis of $\mathbb{R}^n$ and coordinates (3.2A) Quiz LS 10-11
March 17 – 21: No classes. Midterm Break		
March 24 Change of bases (3.2B)	March 26 Subspace defined as a span (3.5A)	March 28 Solution spaces (3.5B) Quiz LS 12-13
March 31 Orthogonal complement (6.2B)	April 2 Orthogonal projections (6.3B)	April 4 Finding orthogonal bases (6.4) Quiz LS 14-15 (last chance LS 7-11)
April 7 Introduction to eigenvectors (4.1)	April 9 Finding eigenvalues (4.2A)	April 11 Finding eigenvectors (4.2B) Quiz LS 16-17
April 14 Diagonalization (4.3)	April 16 <i>Application: discrete dynamical systems revisited</i> (4.4) Quiz LS 18 (last chance LS 12-15)	April 18 No classes: Good Friday
April 21 <i>Application: Markov Chains revisited and Google page rank algorithm</i> (4.5)	April 23 <i>Numeric method: finding eigenvectors numerically</i> (5.2)	April 25 <i>Numeric method: LU factorization</i> (5.1) Quiz LS 19-20
April 28 <i>Application: linear programming models</i> (LP 1)	April 30 <i>Application: geometric solutions of linear programming problems</i> (LP 2) Quiz (last chance LS 16-20)	May 2 <i>Application: the Simplex Method</i> (LP 3)
May 5 No Class – Finals Week	May 7 Quiz LS 1-20 (max 6) 3:00-5:00 p.m.	May 9 Subject to change with notice. V 1.0 Jan 8, 2025